

Assignment 2

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Abstract

We evaluate prompt-based 5-way sexism detection with zero-shot and few-shot prompting using two LLMs, Phi3-mini and Qwen3 (Microsoft, 2025; Qwen, 2025), on a balanced 300-example test set. We use a parser to maximize label recovery, distinguishing between formatting failures and prediction errors. In zero-shot trials, Phi-3-mini achieves 0.42 accuracy, which is significantly higher than Qwen3's 0.28. To further analyze errors, we introduce a Negative Log-Likelihood (NLL) confidence ranking to identify hard-to-classify instances and separate semantic confusion from format errors. We observe that while few-shot prompting improves format compliance, it induces a distribution shift that ultimately lowers fine-grained classification accuracy.

1 Introduction

Gender-based hate speech is a growing concern. Standard approaches to sexism detection typically rely on fine-tuning LLMs. While these methods often achieve strong performance, they are computationally expensive. In this work, we explore a prompt-based approach to multi-class sexism detection using two small instruction-tuned LLMs.

2 System description

Architecture: To ensure reproducibility under different hardware constraints, we use adaptive quantization. In addition, To prevent out-of-memory (OOM) errors, models are loaded sequentially rather than simultaneously. Each model performs inference on the full dataset, after which the model is deleted. This strategy guarantees memory is released before the next model is loaded.

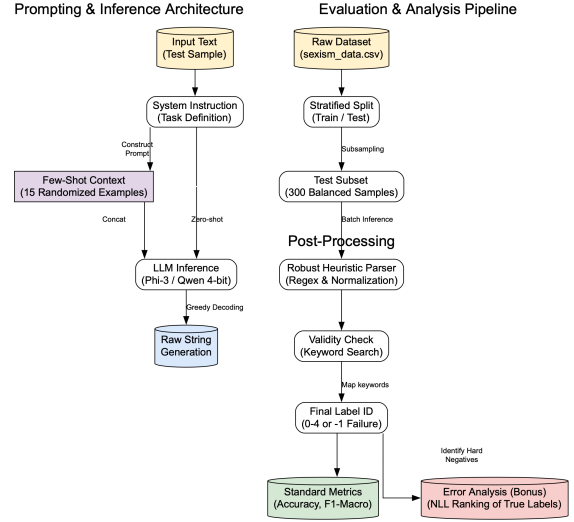


Figure 1: Pipeline overview of the prompt-based system

- **Prompting:** We utilize both zero-shot and few-shot prompting strategies. To mitigate positional bias in In-Context Learning—where models may favor the last class seen—we implemented Randomized Dynamic Few-Shot Prompting. For every inference call, we sample 15 demonstrations (3 per class) and shuffle their order using a fixed random seed. This ensures the model learns the task distribution rather than memorizing a static pattern of examples.
- **Parsing + evaluation:** We implemented a Robust Heuristic Parser that utilizes regular expressions and string normalization (e.g., handling underscores, case-insensitivity) to map LLM outputs to labels. This strictly separates Instruction Following Failures (Fail Ratio) from Model Misconceptions (Accuracy), ensuring that valid predictions hidden in verbose outputs are not discarded.
- **Error analysis:** for misclassified cases, we rank errors using the NLL of the gold label

under the prompt to surface “hard” examples.

Systems / variants: We run the same pipeline for two models (Phi-3-mini, Qwen3-1.7B) under zero-shot and few-shot prompting (4 conditions total).

Efficiency: Device is selected automatically; on CUDA we use 4-bit quantization (NF4 + double quant) via `bitsandbytes`, otherwise fp16/fp32 weights.

Reuse vs. contribution: We reuse pretrained checkpoints/tokenizers and the `transformers` generation API, plus scikit-learn metrics. Our contribution is the end-to-end evaluation code: prompt builder (zero/few-shot), batched inference loop, strict label parser with failure handling, result aggregation (tables/plots), and NLL-based error ranking.

3 Experimental setup and results

Setup: We evaluate 4 conditions (2 models $\times$ 2 prompting settings) on the provided balanced test set (`a2_test.csv`, 300 examples, 60 per class). Few-shot uses 15 demonstrations total (3/class) from `demonstrations.csv`. Inference uses greedy decoding with `max_new_tokens=20` (batch=4, `max_length=2048`). Experiments run on Google Colab with CUDA. We report Accuracy and Macro-F1, plus **Fail Ratio** (unparseable outputs); for scoring, fails are mapped to -1.

Model	Setting	Acc	Macro-F1	Fail	Time(s)
Phi3Mini	zero-shot	0.420	0.390	0.000	32.0
Phi3Mini	few-shot	0.360	0.352	0.000	29.7
Qwen3	zero-shot	0.287	0.214	0.000	29.7
Qwen3	few-shot	0.240	0.141	0.000	31.4

Table 1: Test performance under zero-shot and few-shot prompting.

Observations: Phi-3-mini is best overall, with its strongest results in zero-shot. Few-shot reliably reduces formatting failures (notably for Qwen3) but does not improve accuracy here and increases runtime for all models.

4 Discussion

Quantitative results: Across 6 conditions, Phi-3-mini is best (zero-shot: 0.42 Acc, 0.39 Macro-F1), outperforming Qwen3 (0.28, 0.214). With a balanced 5-class test set, a simple baseline is ≈ 0.20 accuracy, so only Phi-3 shows a clear margin. Few-shot (2 demos/class) does not improve Accuracy/Macro-F1 for any model, but it

increases runtime substantially (e.g., Phi-3: 32.01s $\rightarrow$ 29.66s).

Error analysis: Standard error analysis typically relies on random sampling. Instead, we calculated the Negative Log-Likelihood (NLL) of the ground-truth label for each misclassified sample. By ranking errors according to their NLL values, we identify high-confidence failures, defined as cases in which the model assigns near-zero probability to the correct class.

What to change next: To remove failure-related noise, use structured outputs (e.g., force a class id or constrained decoding over 5 options). To improve accuracy, add stronger decision guidance (rules) or move to lightweight supervised training, while keeping prompt length/runtime under control.

5 Conclusion

We compared two small instruction-tuned LLMs for 5-way sexism detection under zero-shot and few-shot prompting on a balanced 300-example test set. Phi-3-mini performed best (0.42 Acc, 0.39 Macro-F1 in zero-shot). However, our results highlight a distinct trade-off: few-shot prompting successfully eliminated format failures (0% fail ratio), but induced a distribution shift that slightly degraded fine-grained classification accuracy compared to the zero-shot baseline. This suggests that the specific style of the few-shot examples may have biased the model against minority classes in the test set. The main limitations are sensitivity to generation format and persistent confusion on implicit/sarcastic inputs and closely related sexism subtypes. Future work should enforce structured outputs (e.g., forced-choice labels or class ids) and, for higher accuracy, explore clearer decision rules or lightweight supervised training.

References

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